OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION: SALEM

SYSTEM: Emergency Diesel Generator

TASK: Start and synchronize an Emergency Diesel Generator

TASK NUMBER: 113 001 05 01

JPM NUMBER: 15-01 NRC IP-k

ALTERNATE PATH: [ ]

K/A NUMBER: 064 A2.09

IMPORTANCE FACTOR: 3.1 (**RO**) 3.3 (**SRO**)

APPLICABILITY:
EO [ ] **RO** [X] **STA** [ ] **SRO** [X]

EVALUATION SETTING/METHOD: In Plant / Simulate

REFERENCES: 2A Diesel Generator Operation, S2.OP-SO.DG-0001(Q)Rev. 39 (checked 8-5-16)

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 20 min

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By:	**G Gauding** **Instructor**	**Date:** **8-5-16**
Validated By:	**S Bickhart / J Hancock** **SME or Instructor**	**Date:** **9-30-16**
Approved By:	**Training Department**	**Date:** 10/10/16
Approved By:	**Operations Department**	**Date:** 10/10/16

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:
GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

SYSTEM: EDG

TASK: Start and synchronize an Emergency Diesel Generator

TASK NUMBER: 113 001 05 01

INITIAL CONDITIONS: Unit 2 is operating at 100% power, with no equipment OOS. Engineering has requested a loaded run of 2A EDG.

INITIATING CUE:

The Unit 2 CRS has directed you to locally start and parallel load 2A D/G IAW S2.OP-SO.DG-0001(Q), Section 5.2. & 5.4

Prerequisites, Precautions and Limitations, and Section 5.1, Diesel Generator Startup Checks have been performed.

Successful Completion Criteria:

1. **All critical steps completed.**
2. **All sequential steps completed in order.**
3. **All time-critical steps completed within allotted time.**
4. **JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).**

Task Standard for Successful Completion: Perform S2.OP-SO.DG-0001 Sections 5.2 and 5.4 in correct order and to completion.

SYSTEM: Emergency Diesel Generator
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NAME: ______________________
DATE: ______________________

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up copy of S2.OP-SO.DG-0001 2A Diesel Generator Operation			
	5.2.1	**NOTIFY** NCO that 2A Diesel Generator is to be locally started.	Locates nearest page. **Cue:** NCO acknowledges.		
	5.2.2	**CHECK** voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel is OFF	Points out voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel. **Cue:** Light is off		
	5.2.3	**CHECK** speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel is OFF.	Points out speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel. **Cue:** Light is off		
*	5.2.4	IF 2A 4KV Vital Bus is currently energized, (Diesel Generator is to be parallel loaded), THEN: A. **PLACE** 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW in MANUAL (DROOP). B. **ENSURE** B-9, GENERATOR LOADING IN DROOP MODE, is in alarm.	Points out 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW., and simulates rotating to DROOP position. **Cue:** Annunciator B-9, Generator loading in Droop Mode annunciates.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.5	**PLACE** 2A-DF-SS, 2A DG STOP/START SWITCH in START.	Points out 2A-DF-SS, 2A DG STOP/START SWITCH, and simulates placing in START. **Cue:** 2A Diesel is accelerating.		
*	5.2.6	IF Diesel Generator Speed is NOT 900 rpm, THEN **SET** speed to 900 rpm using the SPEED CONTROL SWITCH (GS).	**Cue when Speed indicator is checked:** Speed is 880 rpm. Points out SPEED CONTROL SWITCH (GS) and simulates turning in RAISE direction. **Cue:** 2A Diesel Speed is 900 rpm.		
	5.2.7	IF 2AD1AX6D*, 2A Diesel Generator 125VDC breaker is closed, THEN: • **CHECK** voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel is ON. • **CHECK** speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel is ON.	Points out voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE. **Cue:** Voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE Light is on Points out speed permissive indicator light 2DAE4-LT3, EDG SPEED **Cue:** Speed permissive indicator light 2DAE4-LT3, EDG SPEED light is on		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.2.8	IF Field Ground Relay 64/G white indicating light is OFF, AND C-6, GENERATOR FIELD GROUND, is clear, THEN: A. **RESET** 64/G relay B. **ENSURE** 64/G white indicating light is illuminated.	Points out Field Ground Relay 64/G white indicating light. **Cue:** Field Ground Relay 64/G white indicating light is illuminated.		
	5.2.9	**ENSURE** 2A Diesel Generator K1C Field Flashing Relay Supervisory Light is OFF.	Points out 2A Diesel Generator K1C Field Flashing Relay Supervisory Light **Cue:** 2A Diesel Generator K1C Field Flashing Relay Supervisory Light is off.		
	5.2.10	RECORD the following Diesel Generator Start Readings:	Points out gages and switches to determine readings. **Cue when proper gauge is pointed out :** **2VM189** Gen. Volts - **4160 Volts on all 3 phases** **2FM186** Gen. Frequency - **60 Hz** **2PL6429** LO Hdr. Press - **80 psig** **2PL6449** JW Hdr. Press - **45 psig** **2PL7209** Air Manifold Press **- 0 psig** **2TA16524** Gen. Stator Temp - **187°**		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.2.11	IF 2A Diesel Generator is to be operated unloaded for an extended period of time(>30 minutes), THEN **INITIATE** Section 5.7, Diesel Generator Running Checks.	Determines from initiating cue that 2A EDG is to be parallel loaded.		
	5.2.12	IF Diesel Generator is to be parallel loaded, THEN **INITIATE** Section 5.4, Diesel Generator Parallel Loading.	Initiates Section 5.4, Diesel Generator Parallel Loading.		
	5.4.1	IF Section 5.1 Diesel Generator Startup Checks, was NOT performed, THEN at 2A DG 4KV cabinet cubicle breaker 2AD1AX6D, **PLACE** selector switch 2A DG SYNC ENABLE in ENABLE.	Verifies Section 5.1 is complete.		
	5.4.2	**NOTIFY** NCO that 2A EDG is to be synchronized and loaded locally.	**Cue:** NCO acknowledges 2A DG is to be synchronized and loaded locally.		
	5.4.3	**ENSURE** 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW, in MANUAL (DROOP)	Verifies 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW, is in MANUAL (DROOP)		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.4.4	ADJUST 2A EDG output voltage, as indicated on VOLTMETER-GEN (2VM189) to 50 - 100 Volts higher than that of 2A 4KV Vital Bus voltage, as indicated by VOLTMETER-BUS (2VM190), using the VOLTAGE CONTROL SWITCH (VCS).	**When operator points out 2VM189 and 2VM190, then Cue:** 2VM189 reads 4150, and 2VM190 reads 4150. Points out VOLTAGE CONTROL SWITCH (VCS) and simulates going to RAISE. **Cue:** 2VM189 reads 4210, and 2VM190 reads 4150.		
	5.4.5	**ENSURE** generator terminal voltage is present on all 3 phases by rotating VOLTMETER SWITCH - GEN (VS-G) through each position <u>AND</u> OBSERVING voltmeter 2VM189.	Points out VOLTMETER SWITCH - GEN (VS-G), simulates rotating through each position, while observing 2VM189 **Cue:** Voltage is present on all three phases.		
	5.4.6	**ENSURE** 2A 4KV Vital Bus voltage is present on all 3 phases by rotating VOLTMETER SWITCH - BUS (VS-B) through each position <u>AND</u> OBSERVING voltmeter 2VM190.	Points out VOLTMETER SWITCH - BUS (VS-B), simulates rotating through each position, while observing 2VM190 **Cue:** Voltage is present on all three phases.		

SYSTEM: Emergency Diesel Generator **NAME:** ______________________

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.4.7	PLACE the following switches in "1-2": • VOLTMETER SWITCH - GEN (VS-G) • VOLTMETER SWITCH - BUS (VS-B)	Points out VOLTMETER SWITCH - GEN (VS-G) and VOLTMETER SWITCH - BUS (VS-B) **Cue:** Switches are in 1-2.		
	5.4.8	**ENSURE** 2DAE4-LT2, EDG Voltage indication light is ON.	Points out 2DAE4-LT2, EDG Voltage indication light. **Cue:** 2DAE4-LT2, EDG Voltage indication light is on.		
	5.4.9	**ENSURE** 2DAE4-LT3, EDG Speed indication light is ON.	Points out 2DAE4-LT3, EDG Speed indication light. **Cue:** 2DAE4-LT2, EDG Voltage indication light is on.		
	5.4.10	**SYNCHRONIZE** 2A Diesel Generator to 2A 4KV Vital Bus as follows:			
*	5.4.10 A.	**PLACE** 2A-DF-SYNCH, 2A DG SYNC SWITCH (SS) to ON.	Points out 2A-DF-SYNCH, 2A DG SYNC SWITCH (SS), and simulates placing in ON position.		

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*	5.4.10 B.	**ADJUST** Diesel speed using the SPEED CONTROL SWITCH (GS) such that Synchroscope pointer rotates slowly in the FAST (clockwise) direction.	**When operator points out synchroscope, then Cue:** Scope is going in the slow direction. Points out SPEED CONTROL SWITCH (GS)and simulates going to RAISE. **Cue** Synchroscope pointer rotating slowly in the FAST direction.		
	5.4.10 C.	**ENSURE** the following four SYNC CHECK RELAY 25 UPPER AND LOWER VOLTAGE LIMIT LEDS are ON. "Upper Voltage Limit" "L OK" "B OK" "Lower Voltage Limit" "L OK" "B OK"	Points out four SYNC CHECK RELAY 25 UPPER AND LOWER VOLTAGE LIMIT LEDS. **Cue as each LED is located:** Light is on.		
	5.4.10 D.	<u>IF</u> the SYNC CHK RELAY 25 ΔF OK LED is OFF, <u>THEN</u> **ADJUST** 2A Diesel Generator speed using the SPEED CONTROL SWITCH (GS) until the ΔF OK LED is ON.	Points out SYNC CHK RELAY 25 ΔF OK LED. **Cue** Light is on.		
	5.4.10 E.	ENSURE DG SYNC PERMISSIVE green indicating light is ON each time the synchroscope indicator is near 12 o'clock position (+/- approximately 3 minutes), <u>AND</u> is OFF in any other position of the synchroscope.	**Cue** DG SYNC PERMISSIVE green indicating light is ON each time the synchroscope indicator is near 12 o'clock position (+/- approximately 3 minutes), <u>AND</u> is OFF in any other position of the synchroscope.		

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*	5.4.10 F	**COORDINATE** the following when the synchroscope is at "12 o'clock" (+zero, -2 minutes): 1. **CLOSE** 2A-DF-GCP-3 GENERATOR CIRCUIT BREAKER SWITCH (BCS) 2. **RAISE** 2A Diesel Generator load to ≥ 500KW using SPEED CONTROL SWITCH (GS) to prevent tripping the Diesel Generator Breaker on reverse power.	Points out 2A-DF-GCP-3 GENERATOR CIRCUIT BREAKER SWITCH (BCS) and simulates taking to the CLOSE position when the synch scope is at 12 o'clock. **Cue** 2A Diesel Generator Breaker has closed. Points out SPEED CONTROL SWITCH (GS) simulates going to RAISE while observing KW meter. **Cue** Load is 500 KW.		
	5.4.10 G	**PLACE** 2A-DF-SYNCH, 2A DG SYNCH SWITCH (SS) to OFF.	Points out 2A-DF-SYNCH, 2A DG SYNCH SWITCH (SS), and simulates placing it in OFF position. **Cue:** JPM is complete.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

✓ 1. Task description and number, JPM description and number are identified.

✓ 2. Knowledge and Abilities (K/A) references are included.

✓ 3. Performance location specified. (in-plant, control room, or simulator)

✓ 4. Initial setup conditions are identified.

✓ 5. Initiating and terminating Cues are properly identified.

✓ 6. Task standards identified and verified by SME review.

✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

____ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 39 Date 9/30/16

____ 9. Pilot test the JPM:
   a. verify Cues both verbal and visual are free of conflict, and
   b. ensure performance time is accurate.

____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: Hancock Date: 9/30/16

SME/Instructor: Bickhart Date: 9/30/16

SME/Instructor: ____ Date: ____

INITIAL CONDITIONS:

Unit 2 is operating at 100% power, with no equipment OOS. Engineering has requested a loaded run of 2A EDG.

INITIATING CUE:

The Unit 2 CRS has directed you to locally start and parallel load 2A D/G IAW S2.OP-SO.DG-0001(Q), Section 5.2. & 5.4

Prerequisites, Precautions and Limitations, and Section 5.1, Diesel Generator Startup Checks have been performed.